



# **Experimental methods in solid-state physics**

## **04 Basic theory of diffraction**

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und Nanomaterialien



Max von Laue  
Nobel Prize in Physics 1914

Born: 9 October 1879, Pfaffendorf, Germany

Died: 23 April 1960, Berlin, West Germany (now Germany)

Affiliation at the time of the award: Frankfurt-on-the-Main University, Frankfurt-on-the-Main, Germany

Prize motivation: “for his discovery of the diffraction of X-rays by crystals”

Prize share: 1/1

Photo: Franz Schmelhaus.  
Nobel Foundation archive

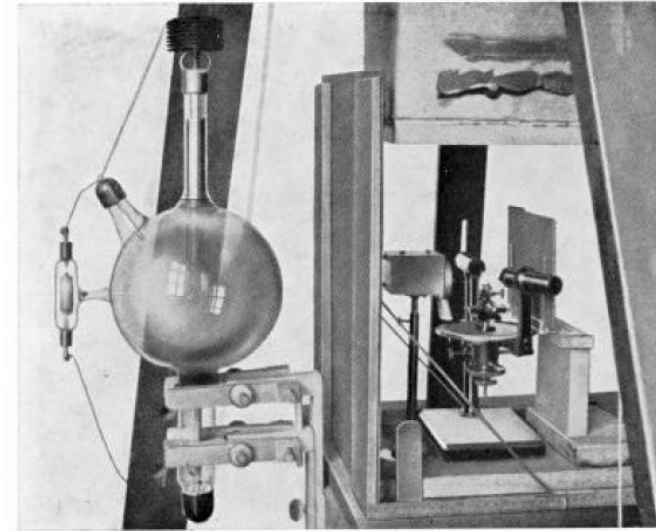
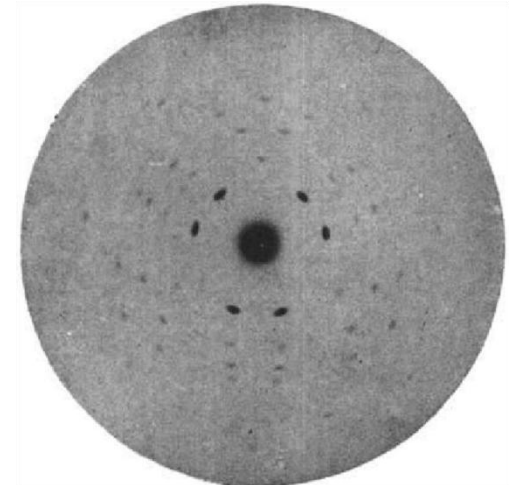
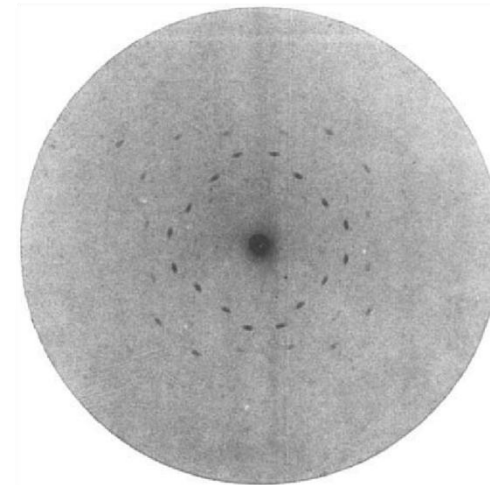


Fig. 4-4(2). Friedrich & Knipping's improved set-up.





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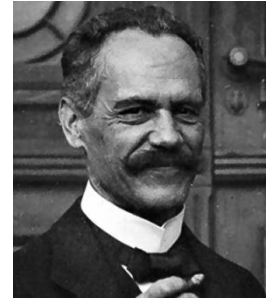
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Experimental work carried out at  
chair of Arnold Sommerfeld  
(LMU, theoretical physics)



3. *Interferenzerscheinungen bei Röntgenstrahlen;  
von W. Friedrich, P. Knipping und M. Laue.*<sup>1)</sup>

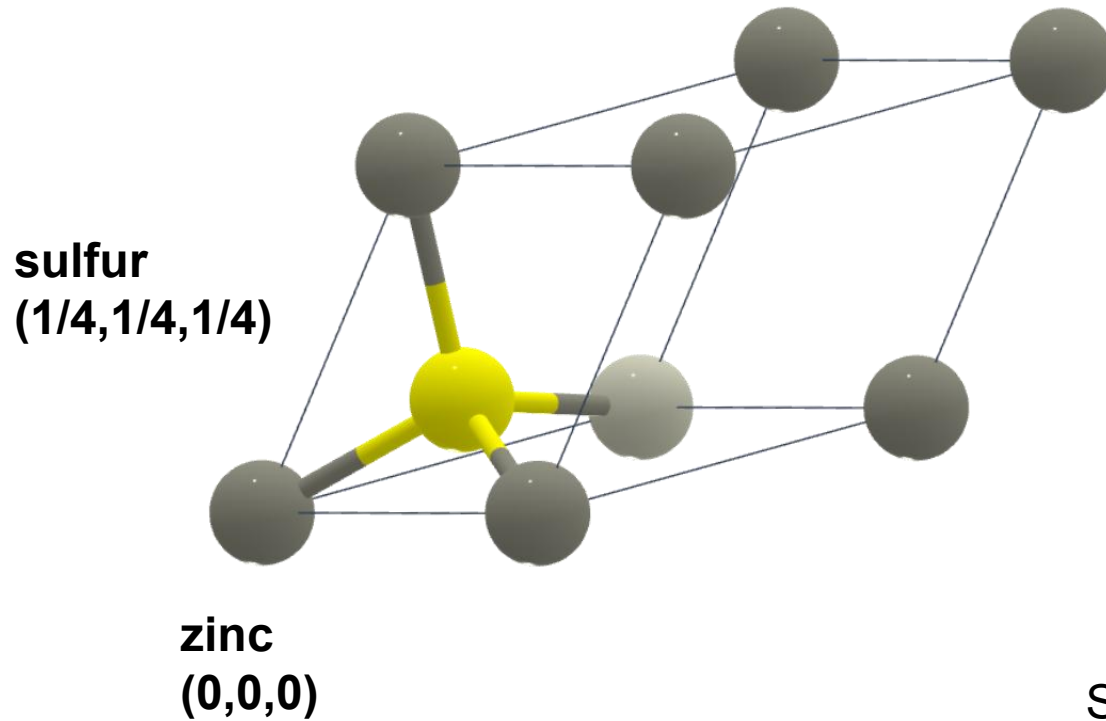
(Hierzu Taf. I—IV, Figg. 1—10.)

Zum Schluß wollen wir besonders Hrn. Prof. A. Sommerfeld, in dessen Institut diese Arbeit ausgeführt ist, für die bereitwillige Beschaffung der Apparate, sowie sein stetiges Interesse und seine fördernden Ratschläge unseren herzlichsten Dank aussprechen. Wir möchten bei dieser Gelegenheit auch nicht versäumen, Hrn. Geheimrat Röntgen und Hrn. Geheimrat v. Groth für Überlassung von wertvollen Kristallen und Apparaten, sowie ihr reges Interesse ebenfalls verbindlich zu danken.

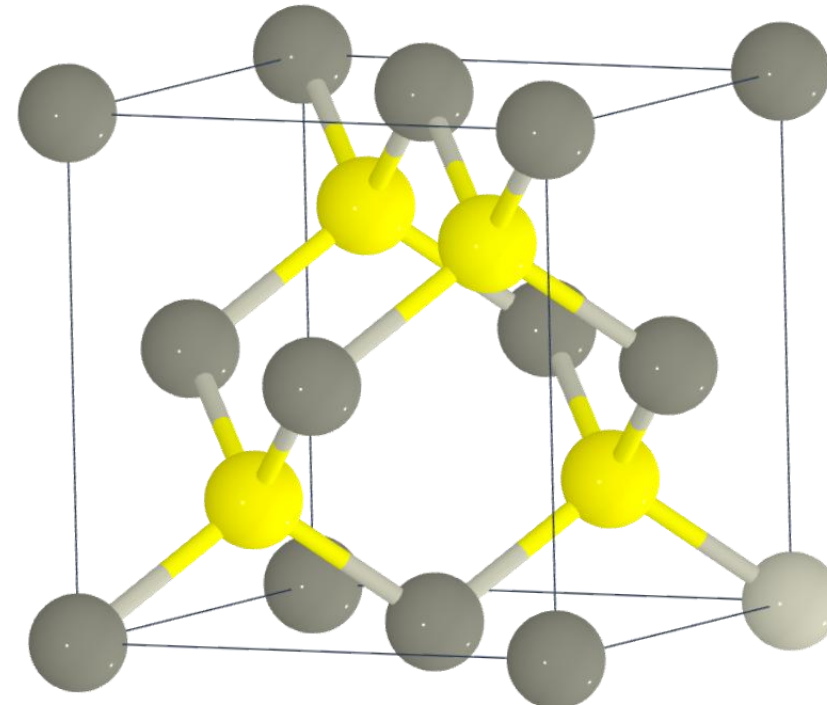
München, Inst. für theor. Physik der Universität.

(Eingegangen 15. März 1913.)

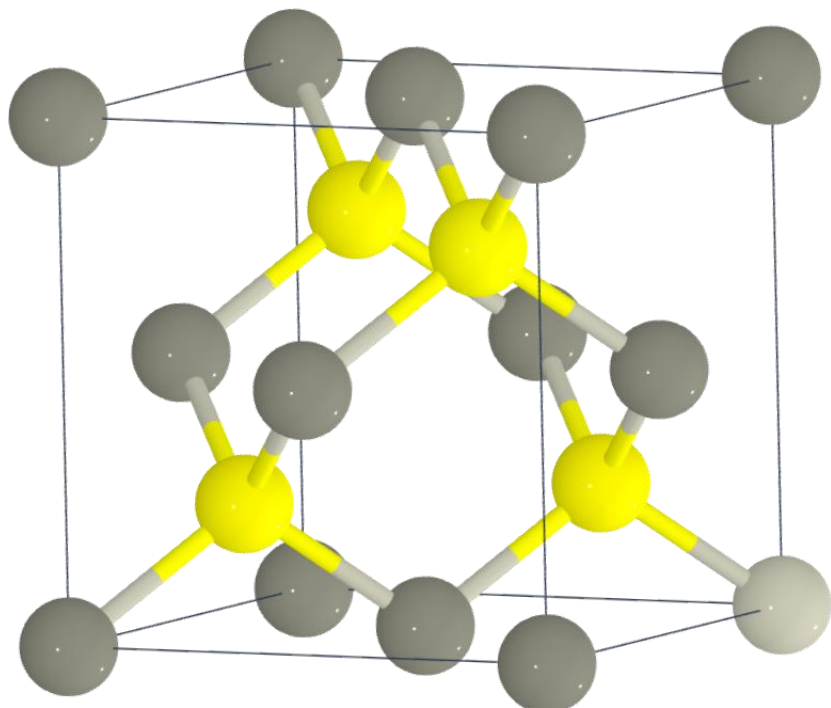
Primitive unit cell



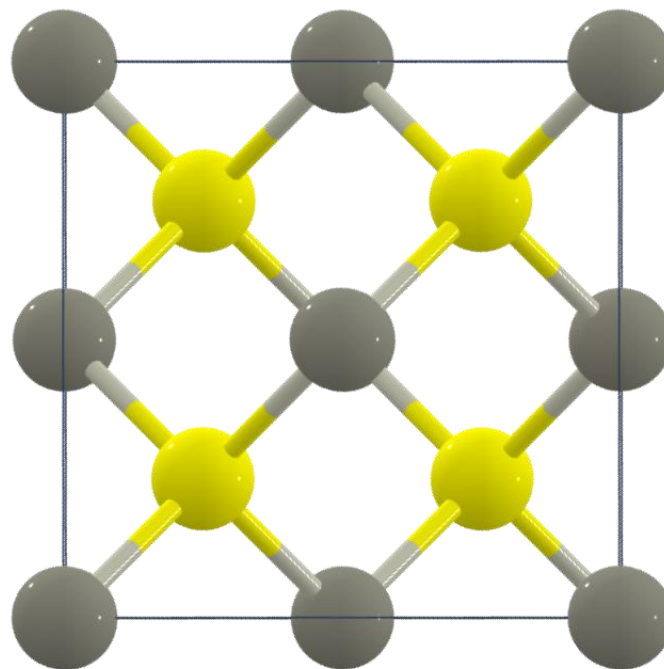
Conventional unit cell



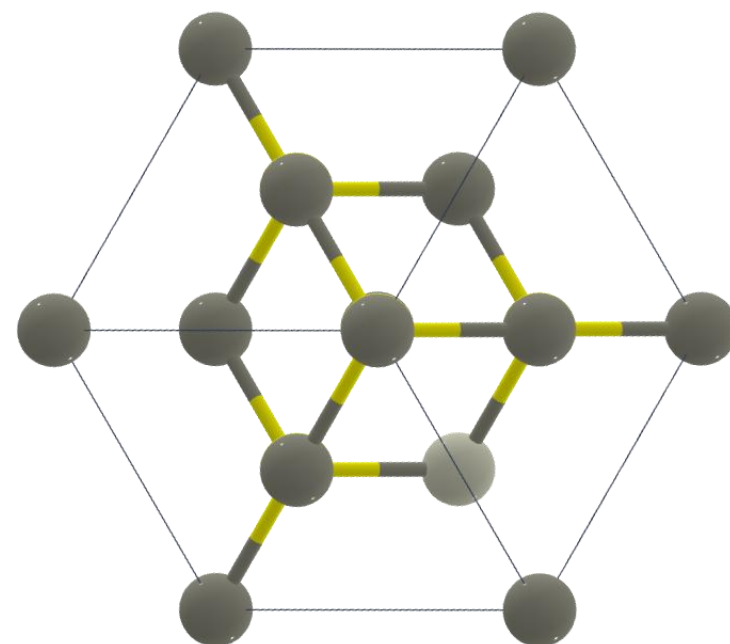
Same lattice GaAs (III-V) or CdTe (II-VI), but also silicon and diamond (two identical atoms in the basis)



**[100]**  
„the sides of the cube“



**[111]**  
„the diagonal of the cube“





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 **Materials Explorer** 📄 References 📄 Documentation

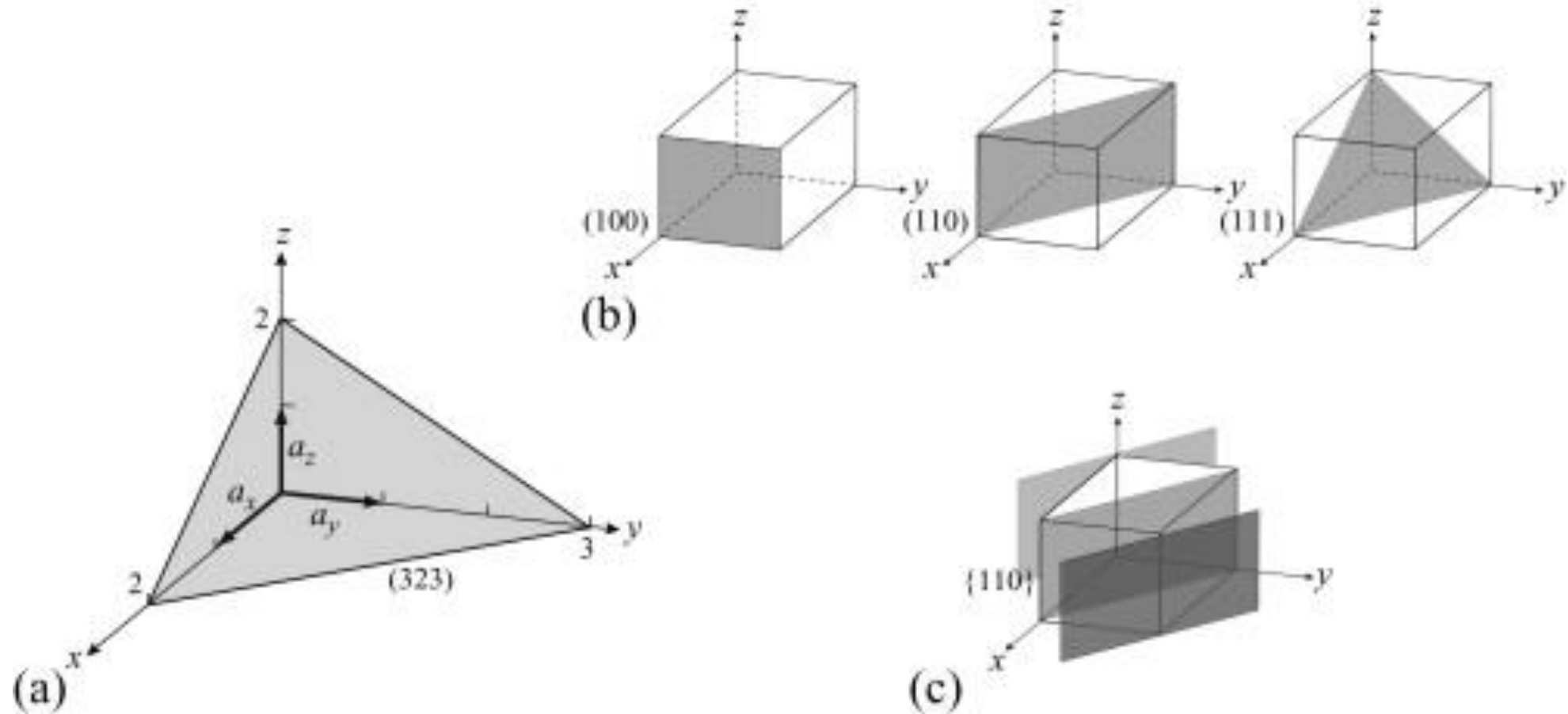
Search for materials information by chemistry, composition, or property.

Materials e.g. Li-Fe or Li,Fe or Li3Fe or mp-19017 ? 📄 Search

**Only Elements** At Least Elements Formula

\* Select elements to search for materials with **only** these elements

|    |    |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |  |    |
|----|----|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--|--|--|----|
| H  |    |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |  | He |
| Li | Be |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |  |    |
| Na | Mg |       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |  |    |
| K  | Ca | Sc    | Ti | V  | Cr | Mn | Fe | Co | Ni | Cu | Zn | Ga | Ge | As | Se | Br | Kr |  |  |  |    |
| Rb | Sr | Y     | Zr | Nb | Mo | Tc | Ru | Rh | Pd | Ag | Cd | In | Sn | Sb | Te | I  | Xe |  |  |  |    |
| Cs | Ba | La-Lu | Hf | Ta | W  | Re | Os | Ir | Pt | Au | Hg | Tl | Pb | Bi | Po | At | Rn |  |  |  |    |
| Fr | Ra | Ac-Lr | Rf | Db | Sg | Bh | Hs | Mt | Ds | Rg | Cn | Nh | Fl | Mc | Lv | Ts | Og |  |  |  |    |
|    |    |       | La | Ce | Pr | Nd | Pm | Sm | Eu | Gd | Tb | Dy | Ho | Er | Tm | Yb | Lu |  |  |  |    |
|    |    |       | Ac | Th | Pa | U  | Np | Pu | Am | Cm | Bk | Cf | Es | Fm | Md | No | Lr |  |  |  |    |





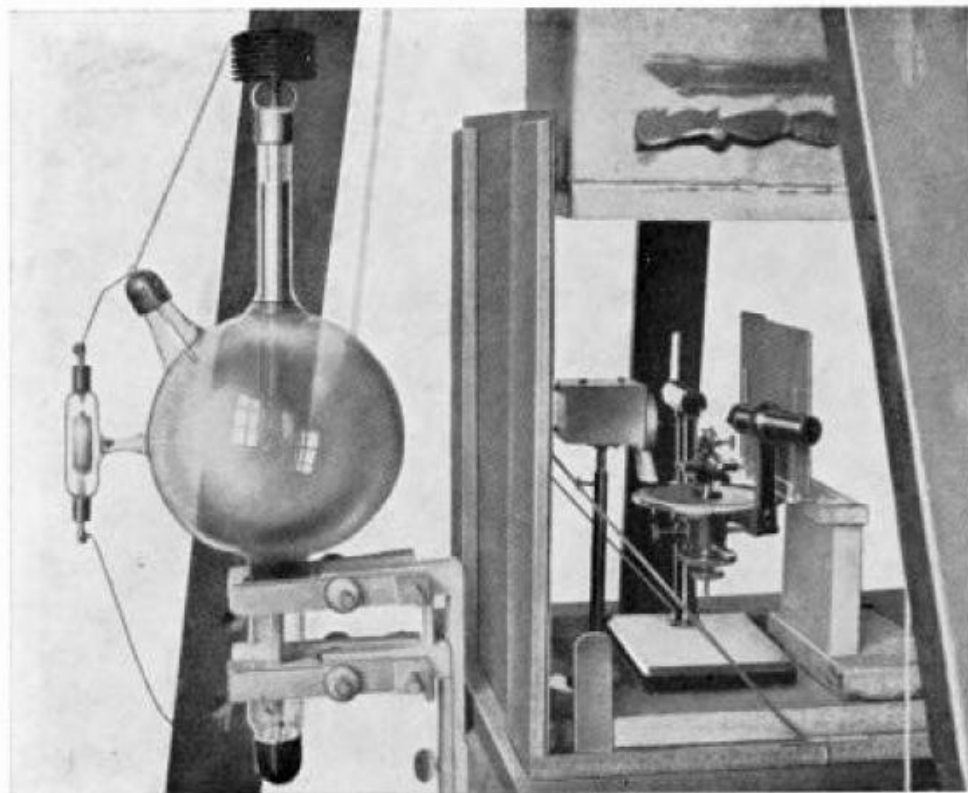
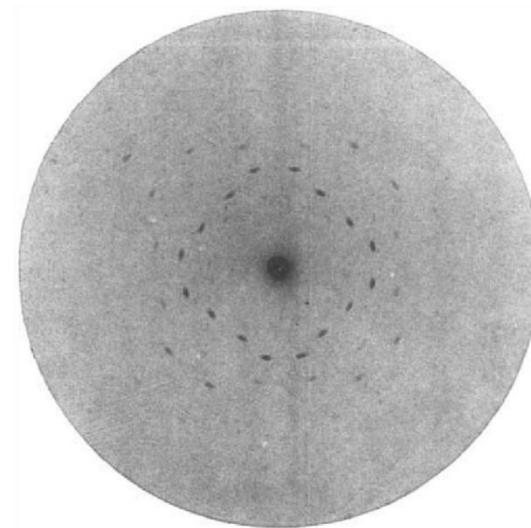
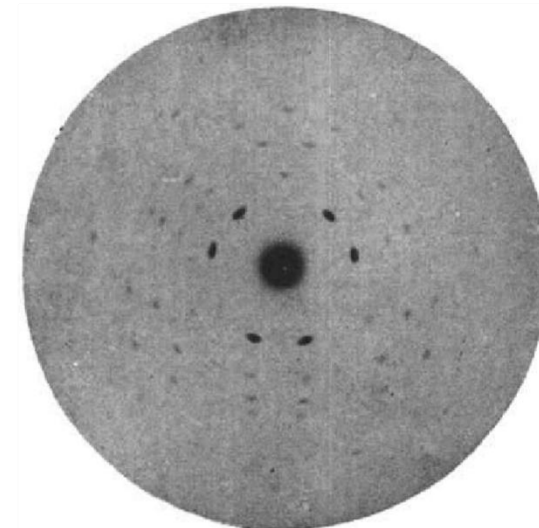


Fig. 4-4(2). Friedrich & Knipping's improved set-up.

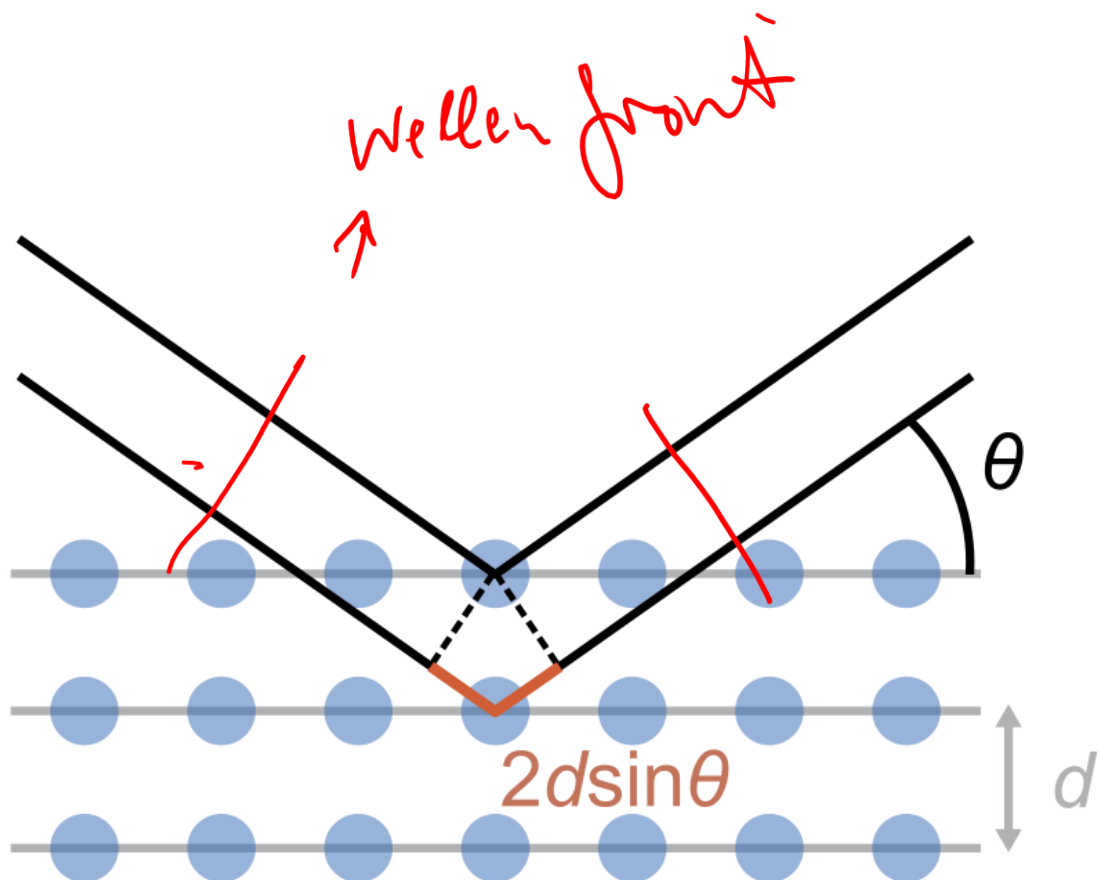


X-rays incident along [100]



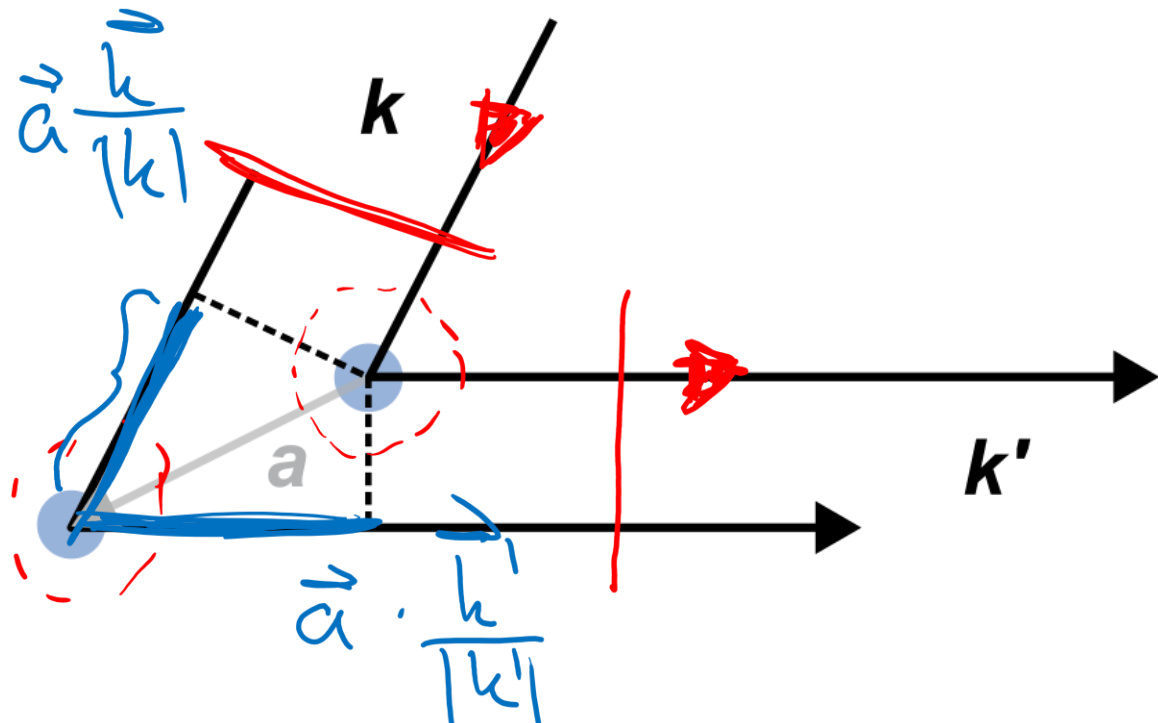
[111]





konstruktiv  
 $\Delta s = n \cdot \lambda$

$$2d \sin \theta = n \cdot \lambda$$



„Gross“-convention:

- Incoming wave vector  $\mathbf{k}$
- Outgoing wave vector  $\mathbf{k}'$

„Hunklinger“-convention:

- Incoming wave vector  $\mathbf{k}_0$
- Outgoing wave vector  $\mathbf{k}$

$$\Delta s = \vec{a} \cdot \frac{-\vec{k}}{|\vec{k}|} + \vec{a} \cdot \frac{\vec{k}'}{|\vec{k}'|}$$

$$= \vec{a} \cdot \left( \frac{\vec{k}'}{|\vec{k}'|} - \frac{\vec{k}}{|\vec{k}|} \right)$$

elastisch  $|\vec{k}'| = |\vec{k}|$

$$\vec{a} \cdot \underbrace{\left( \frac{\vec{k}'}{|\vec{k}'|} - \frac{\vec{k}}{|\vec{k}|} \right)}_{\Delta \vec{k}} = \frac{\vec{a} \cdot \Delta \vec{k}}{2n} \Delta \vec{k} = n\lambda$$



## Reciprocal Lattice

The reciprocal lattice provides a compact description of the periodicity of a crystal in wavevector space. It follows directly from the Laue interference condition  $e^{i\Delta\mathbf{k}\cdot\mathbf{a}_i} = 1$ , which implies  $\Delta\mathbf{k} \cdot \mathbf{a}_i = 2\pi n_i$  for all primitive vectors  $\mathbf{a}_i$ .

**Definition.** The reciprocal basis vectors  $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$  are defined by

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi \delta_{ij}.$$

Any reciprocal-lattice vector is

$$\mathbf{G}_{hkl} = h \mathbf{b}_1 + k \mathbf{b}_2 + \ell \mathbf{b}_3, \quad h, k, \ell \in \mathbb{Z}.$$

Each  $\mathbf{G}_{hkl}$  corresponds to a periodic modulation that repeats in phase after every lattice translation.

**Explicit construction.** For a direct lattice spanned by  $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$  with cell volume  $V = \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)$ ,

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{V}, \quad \mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{V}, \quad \mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{V}.$$

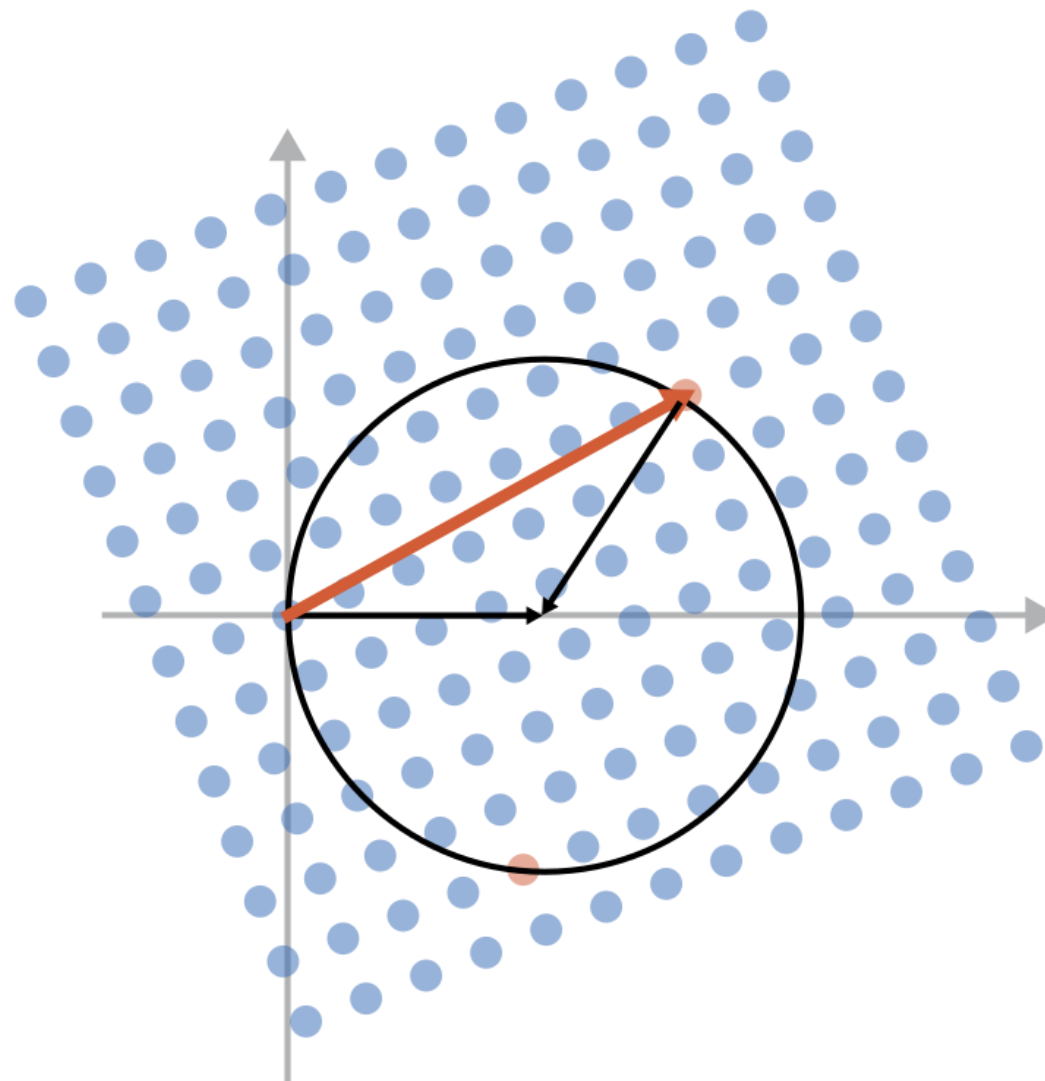
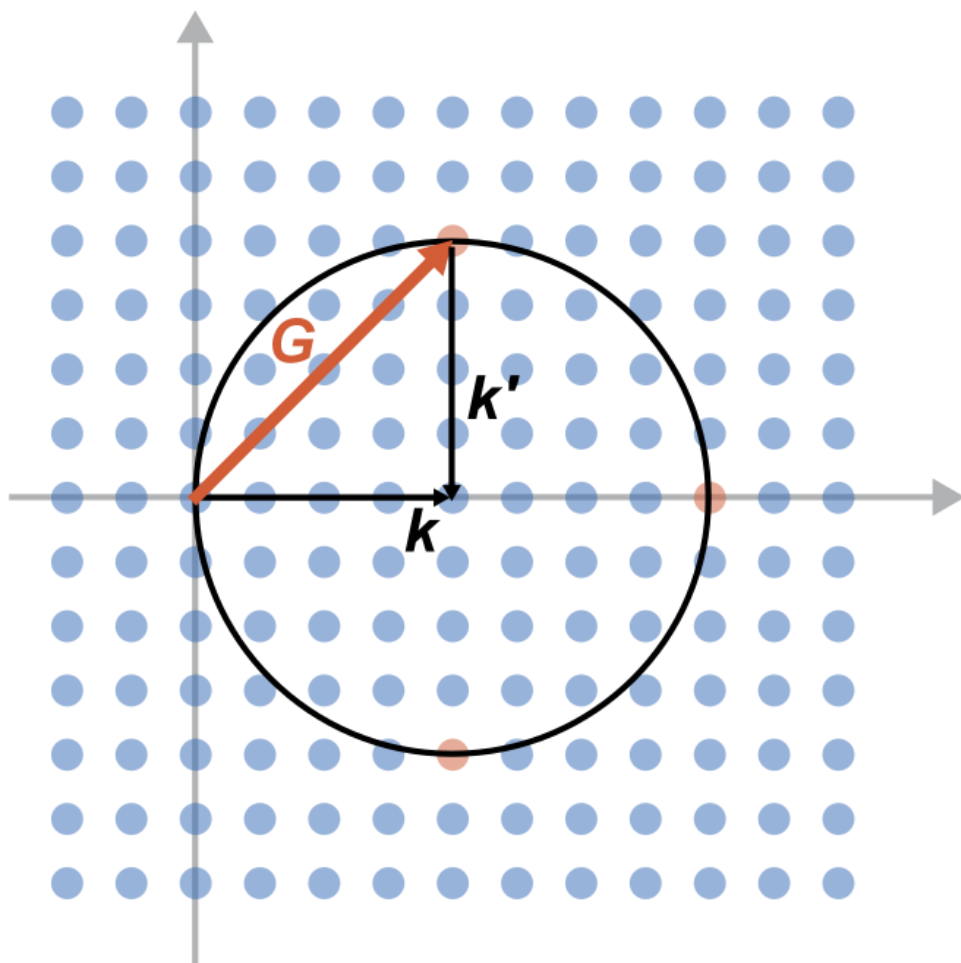
The reciprocal cell volume is  $V^* = (2\pi)^3/V$ .

**Geometric meaning.** Each  $\mathbf{G}_{hkl}$  is perpendicular to the family of planes  $(hkl)$ , satisfying  $\mathbf{G}_{hkl} \cdot \mathbf{r} = 2\pi n$ . The spacing between adjacent planes is  $d_{hkl} = 2\pi/|\mathbf{G}_{hkl}|$ . In cubic crystals,  $\mathbf{b}_i$  are parallel to  $\mathbf{a}_i$  with  $|\mathbf{b}_i| = 2\pi/a$ .

**Diffraction condition.** Constructive interference (Laue condition) occurs when

$$\mathbf{k}' - \mathbf{k} = \mathbf{G}_{hkl}.$$

Thus, each diffraction peak corresponds to one reciprocal-lattice vector lying on the Ewald sphere of radius  $|\mathbf{k}| = 2\pi/\lambda$ .

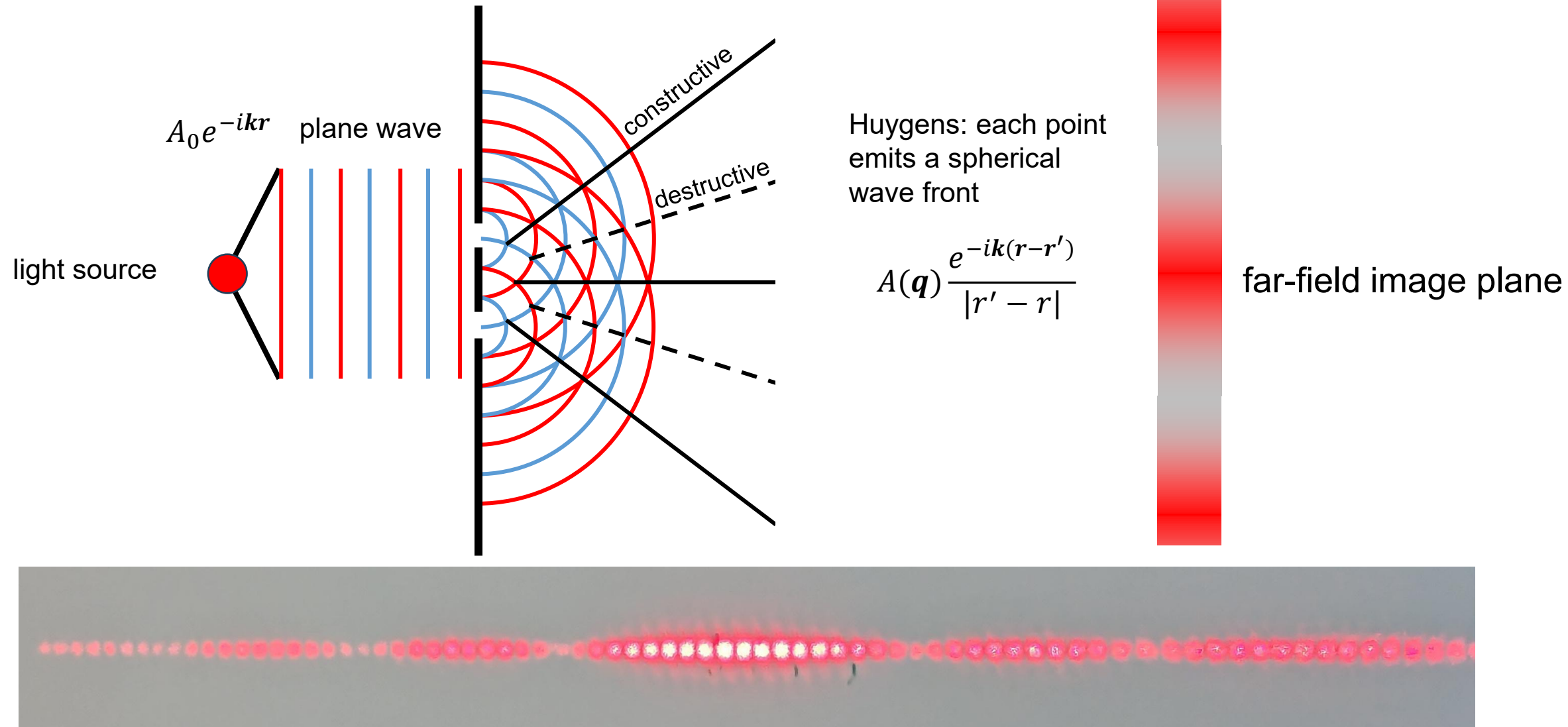




# **Experimental methods in solid-state physics**

## **05 Scattering as Fourier transform: Form factor and structure factor**

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## Fraunhofer-Näherung [ [Bearbeiten](#) | [Quelltext bearbeiten](#) ]

Die **Fraunhofer-Näherung** entspricht einer *Fernfeld*-Näherung, das heißt, dass nicht nur die Blendenöffnung als klein, sondern auch die Entfernung des Beobachtungsschirms  $L$  als groß angenommen werden. Als Beugungsintegral ergibt sich dabei im Wesentlichen gerade die *Fourier-Transformierte* der Blendenfunktion. Deshalb spricht man im Rahmen der Fraunhofer-Beugung auch von der *Fourier-Optik*.

Entsprechend diesen Annahmen werden nur Terme berücksichtigt, die linear in  $x$  und  $y$  sind, das heißt

$$d \approx L \left( 1 + \frac{x'^2 + y'^2 - 2(x x' + y y')}{2L^2} \right) = L \left( 1 + \frac{|\vec{p}|^2 - 2(x x' + y y')}{2L^2} \right),$$
$$d_1 \approx L_1.$$

In diesem Fall vereinfacht sich das Beugungsintegral zu

$$\psi_P \approx \frac{a_Q k_0}{2\pi i} \frac{e^{i k_0 \left( L_1 + L + \frac{p^2}{2L} \right)}}{L_1 L} \iint_{\text{Blende}} dx dy f_S(x, y) e^{-i k_0 \frac{(x x' + y y')}{L}}.$$

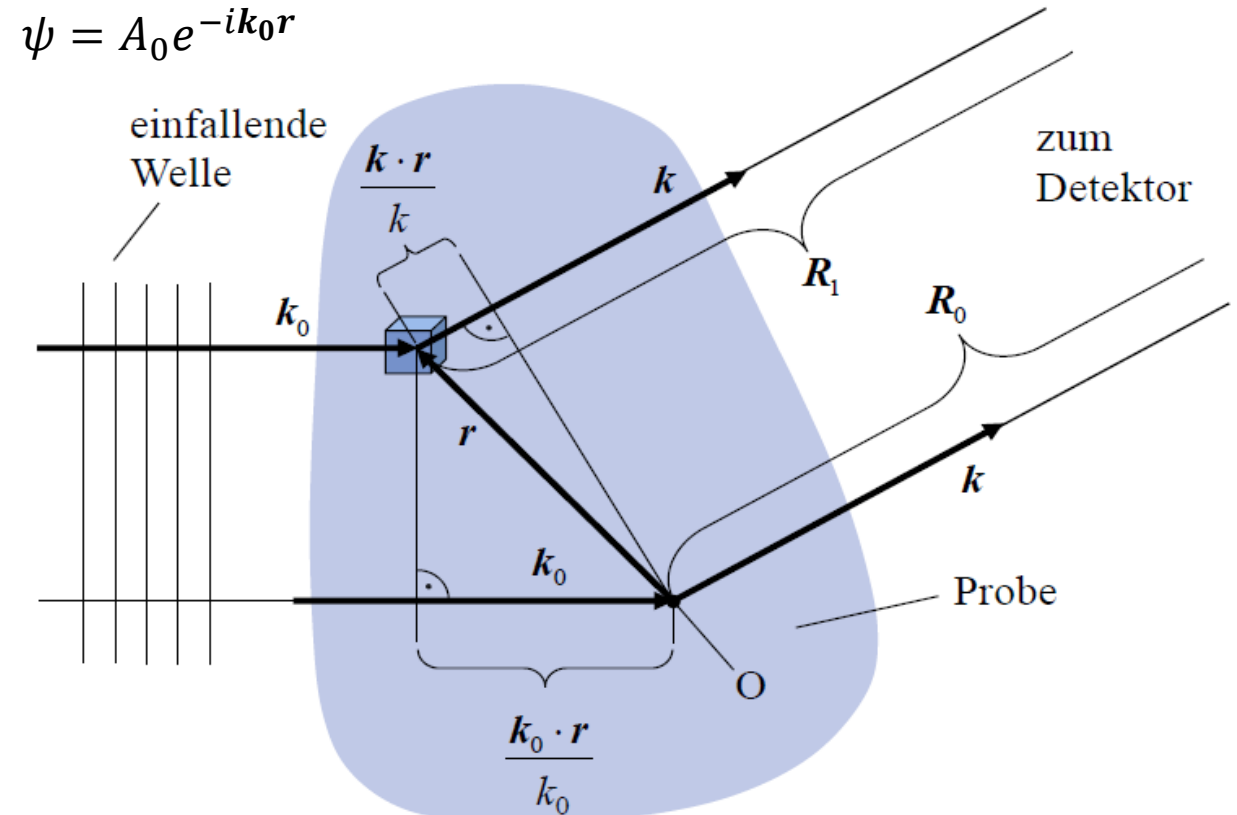
Definiert man einen neuen Wellenvektor  $\vec{K} = \frac{k_0}{L} \vec{p}$ , so ergibt sich für das Integral

$$\iint_{\text{Blende}} dx dy f_S(x, y) e^{-i k_0 \frac{(x x' + y y')}{L}} = \iint_{\text{Blende}} dx dy f_S(x, y) e^{-i \frac{k_0}{L} \vec{p} \cdot \vec{s}} = \iint_{\text{Blende}} dx dy f_S(x, y) e^{-i \vec{K} \cdot \vec{s}}.$$

Dies ist gerade die Fourier-Transformierte der Blendenfunktion  $f_S$ .

## Assumption: scattering is weak

- less than one scattering event inside the sample volume
- approximate the local solution inside the material by the incoming wave
- equivalent to first-order perturbation theory
- or equivalent to calculating only the first Feynman diagram



Hunklinger, Festkörperphysik



# Basic scattering theory and Born approximation



**1. From Schrödinger to the inhomogeneous Helmholtz equation.** For a particle of mass  $m$  in a scattering potential  $V(\mathbf{r})$ , the stationary Schrödinger equation can be rewritten as an inhomogeneous Helmholtz equation

$$(\nabla^2 + k^2)\psi(\mathbf{r}) = \frac{2m}{\hbar^2} V(\mathbf{r}) \psi(\mathbf{r}), \quad k^2 = \frac{2mE}{\hbar^2},$$

**2. Outgoing Green function.** The Green function of  $(\nabla^2 + k^2)$  is defined by

$$(\nabla^2 + k^2) G^{(+)}(\mathbf{r}, \mathbf{r}') = -\delta(\mathbf{r} - \mathbf{r}').$$

The physical (outgoing) solution is a spherical wave

$$G^{(+)}(\mathbf{r}, \mathbf{r}') = -\frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|},$$

**3. Lippmann–Schwinger equation.** Multiply the Helmholtz equation for  $\psi$  by  $G^{(+)}$ , the Green equation by  $\psi$ , subtract, and integrate over all space. Using Gauss's theorem and outgoing boundary conditions yields the integral form of the scattering state:

$$\psi(\mathbf{r}) = \psi_{\text{in}}(\mathbf{r}) + \frac{2m}{\hbar^2} \int G^{(+)}(\mathbf{r}, \mathbf{r}') V(\mathbf{r}') \psi(\mathbf{r}') d^3r'.$$

This expresses the scattered wave as a superposition of outgoing spherical waves emitted from each point in the potential.



# Basic scattering theory and Born approximation

$$\psi(\mathbf{r}) \approx e^{i\mathbf{k}\cdot\mathbf{r}} + \frac{2m}{\hbar^2} \int G^{(+)}(\mathbf{r}, \mathbf{r}') V(\mathbf{r}') e^{i\mathbf{k}\cdot\mathbf{r}'} d^3r'.$$

**5. Far-field limit and scattering amplitude.** For  $|\mathbf{r}| \rightarrow \infty$ ,

$$G^{(+)}(\mathbf{r}, \mathbf{r}') \approx -\frac{e^{ikr}}{4\pi r} e^{-i\mathbf{k}'\cdot\mathbf{r}'}, \quad \mathbf{k}' = k \hat{\mathbf{r}}.$$

Thus the scattered wave has the standard asymptotic form

$$\psi(\mathbf{r}) \xrightarrow{r \rightarrow \infty} e^{i\mathbf{k}\cdot\mathbf{r}} + f(\mathbf{k} \rightarrow \mathbf{k}') \frac{e^{ikr}}{r},$$

with the Born amplitude

$$f(\mathbf{k} \rightarrow \mathbf{k}') = -\frac{2m}{4\pi\hbar^2} \int V(\mathbf{r}) e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} d^3r.$$

**6. Application to X-ray scattering.** For X-rays, interaction is governed by  $V(\mathbf{r}) \propto \epsilon(\mathbf{r}) \propto \rho(\mathbf{r})$ . Therefore, we define the scattering amplitude as

$$A(\mathbf{q}) = \int \rho(\mathbf{r}) e^{-i\mathbf{q}\cdot\mathbf{r}} d^3r, \quad \mathbf{q} = \mathbf{k}' - \mathbf{k},$$

where the integral runs across the sample volume, i.e. the scattered amplitude is the Fourier transform of the electron density.



# *Atomic form factor*

## Density at atomic positions

$$\rho(\mathbf{r}) = \sum_{\alpha} \rho_{\alpha}(\mathbf{r} - \mathbf{r}_{\alpha}),$$

## Scattering amplitude

$$A(\mathbf{q}) = \int \rho(\mathbf{r}) e^{-i\mathbf{q} \cdot \mathbf{r}} d^3r,$$

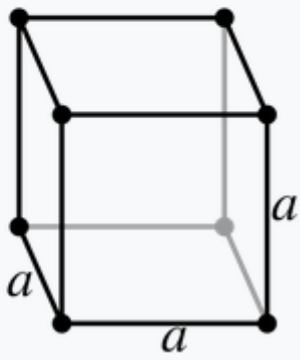
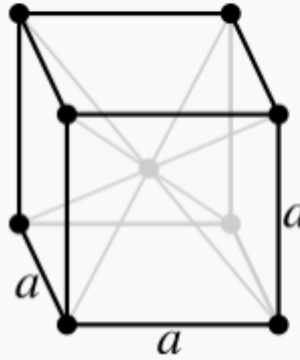
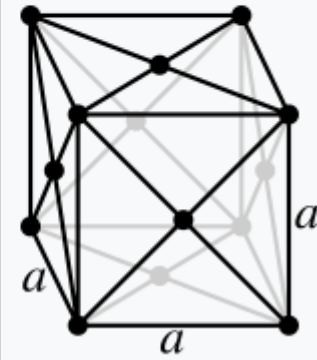


Density at **periodic** atomic positions

$$\rho(\mathbf{r}) = \sum_{\mathbf{R}} \sum_{\alpha} \rho_{\alpha}(\mathbf{r} - \mathbf{R} - \mathbf{r}_{\alpha}),$$

Scattering amplitude

$$A(\mathbf{q}) = \sum_{\mathbf{R}} \sum_{\alpha} f_{\alpha}(\mathbf{q}) e^{-i\mathbf{q} \cdot (\mathbf{R} + \mathbf{r}_{\alpha})}.$$

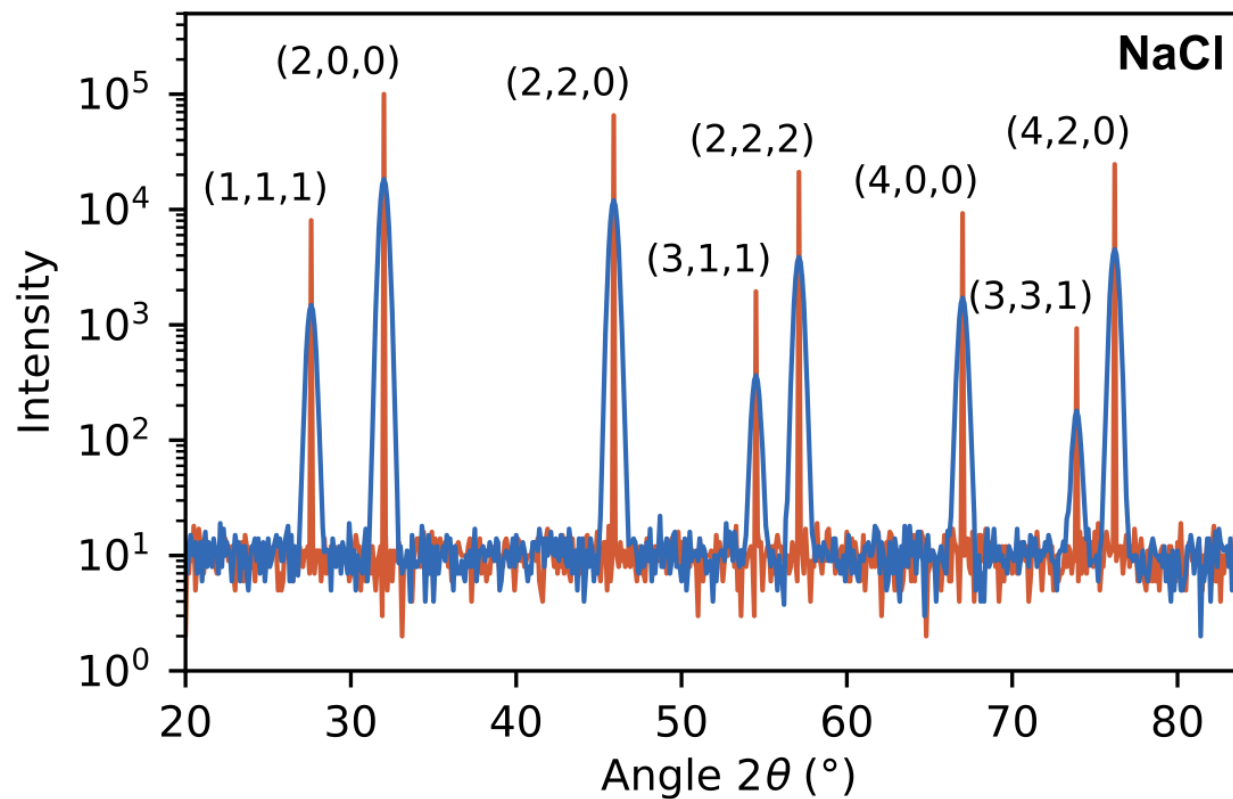
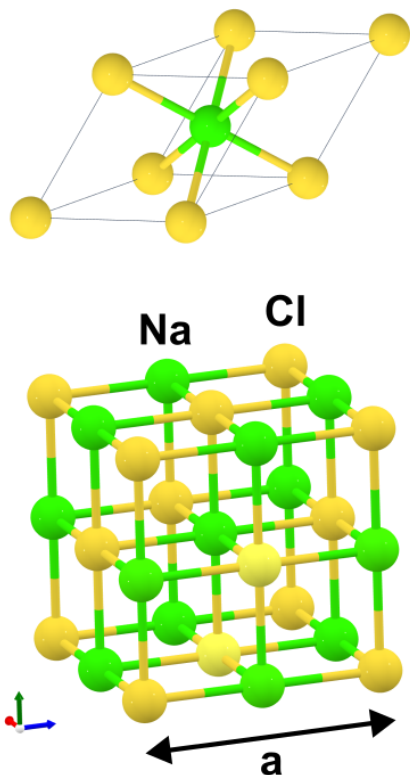
| Bravais lattice        | Primitive cubic                                                                    | Body-centered cubic                                                                 | Face-centered cubic                                                                  |
|------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Pearson symbol         | $cP$                                                                               | $cI$                                                                                | $cF$                                                                                 |
| Conventional Unit cell |  |  |  |

- The reference is always the primitive cubic system.
- All diffraction peak are indexed as (hkl) where hkl denote planes in the primitive cubic basis.

Why: 
$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

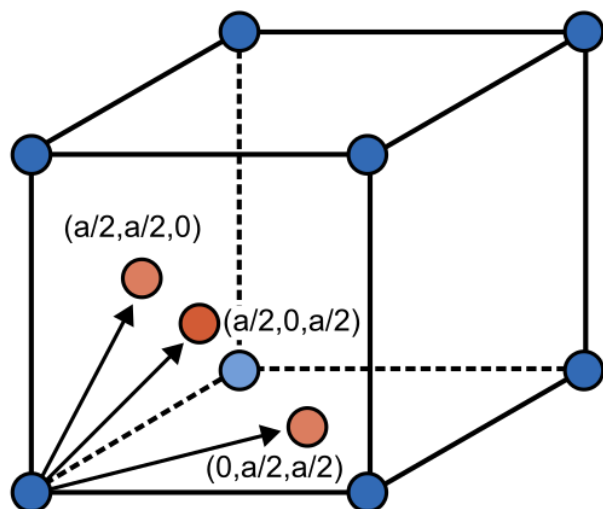
$$\sin^2(\theta) = \frac{\lambda^2}{4a^2} (h^2 + k^2 + l^2)$$

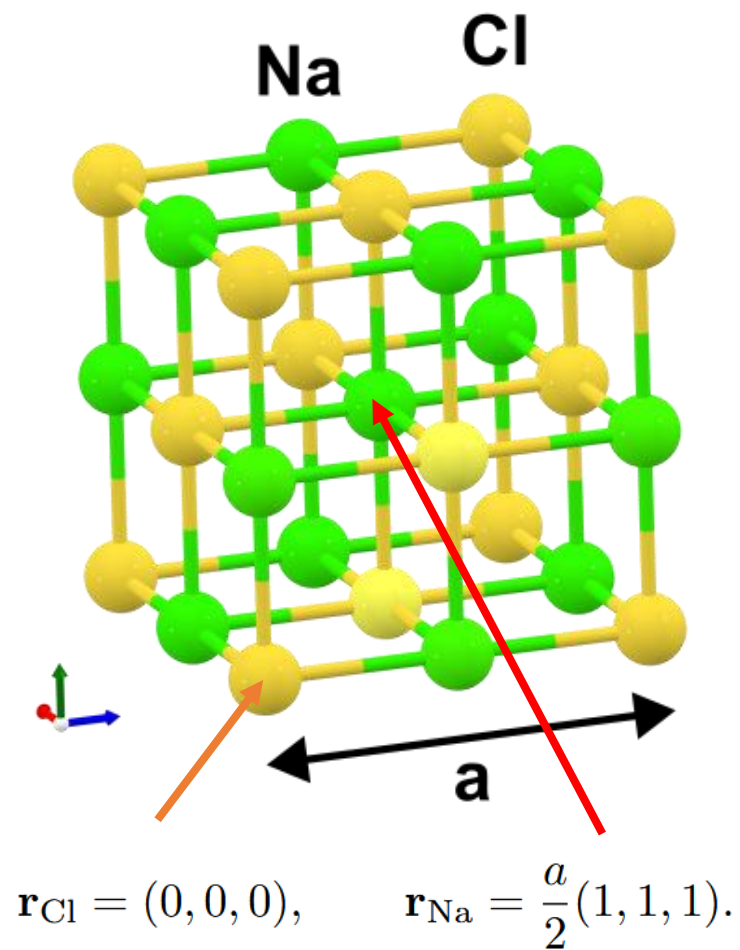
## Example of NaCl



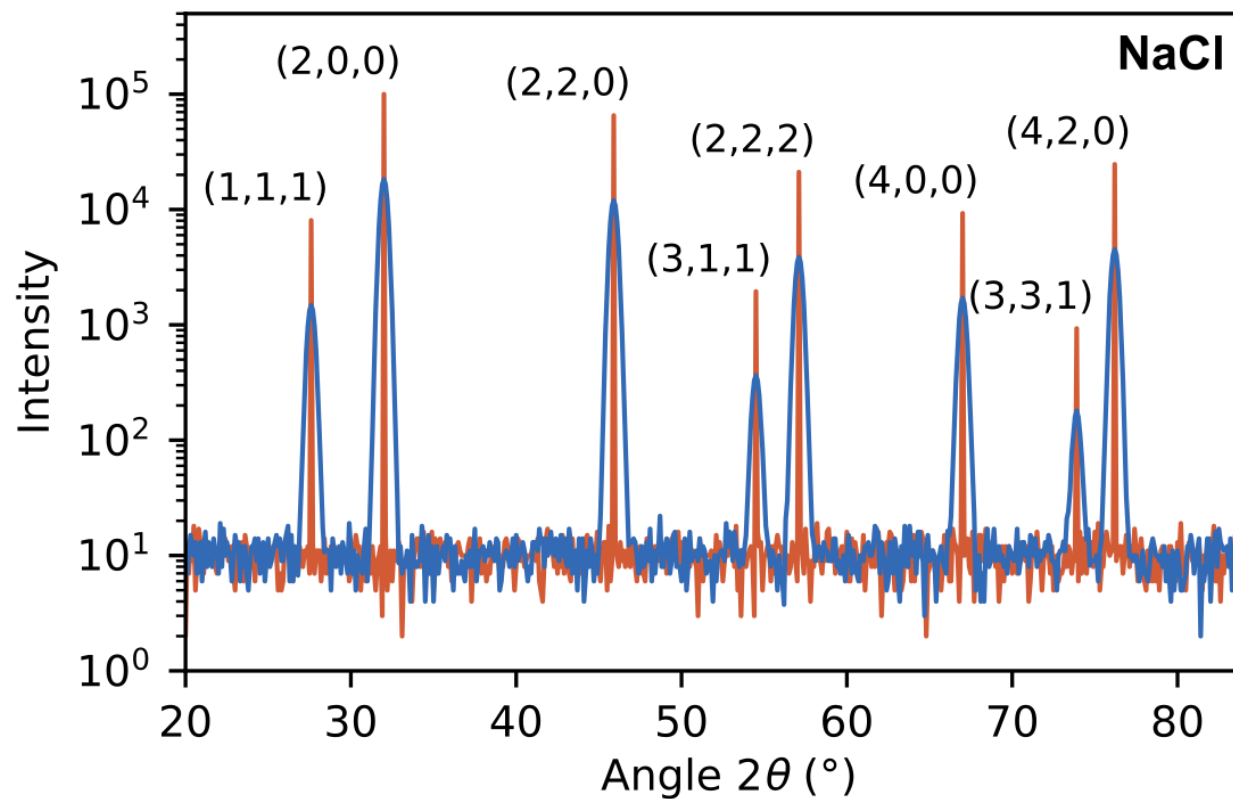
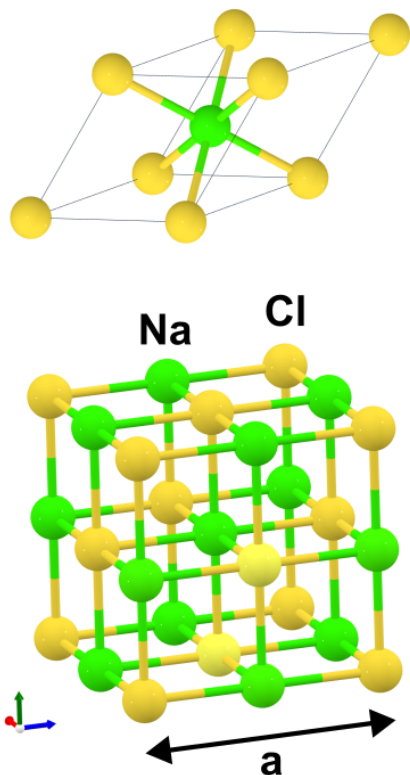
- fcc lattice with 2 atoms in the basis
- hkl all even or all odd
- hkl even  $\rightarrow$  strong
- hkl odd  $\rightarrow$  weak



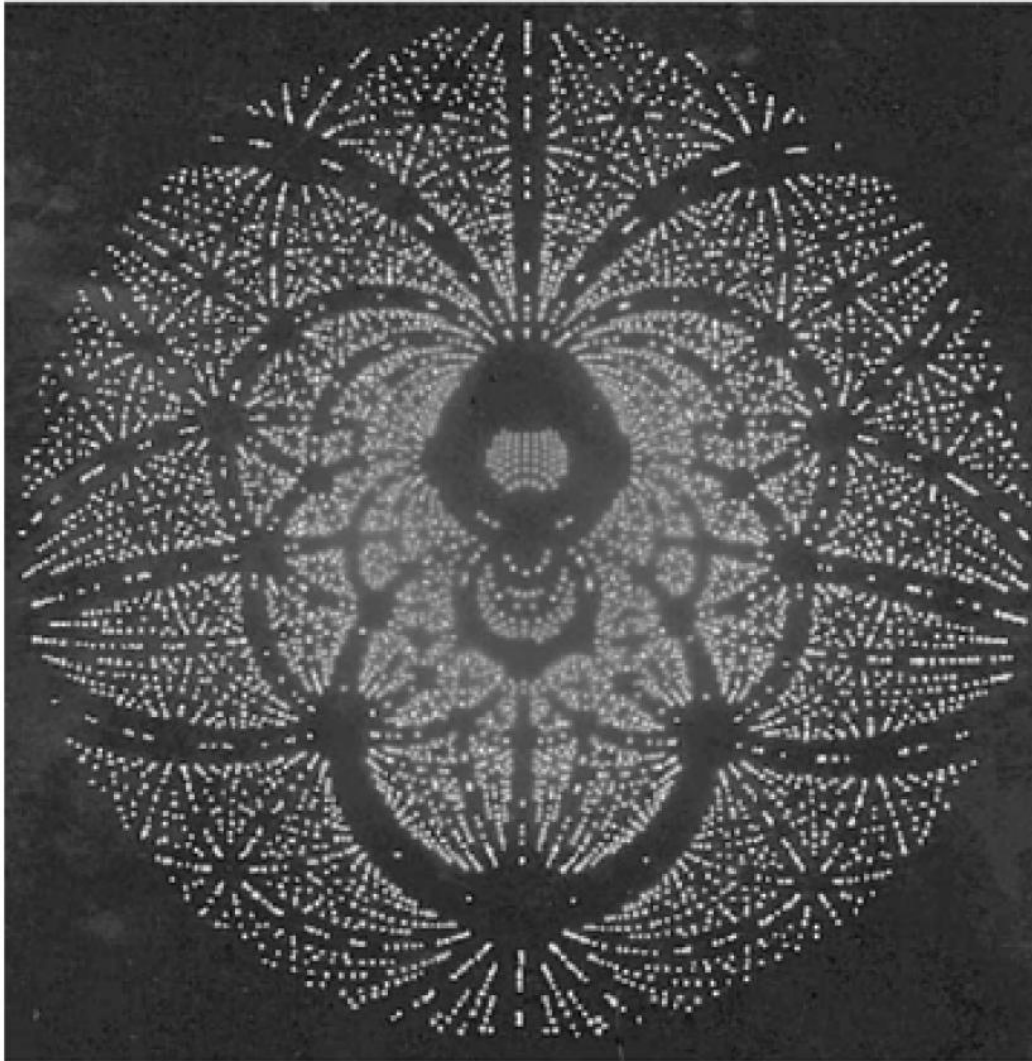




## Example of NaCl

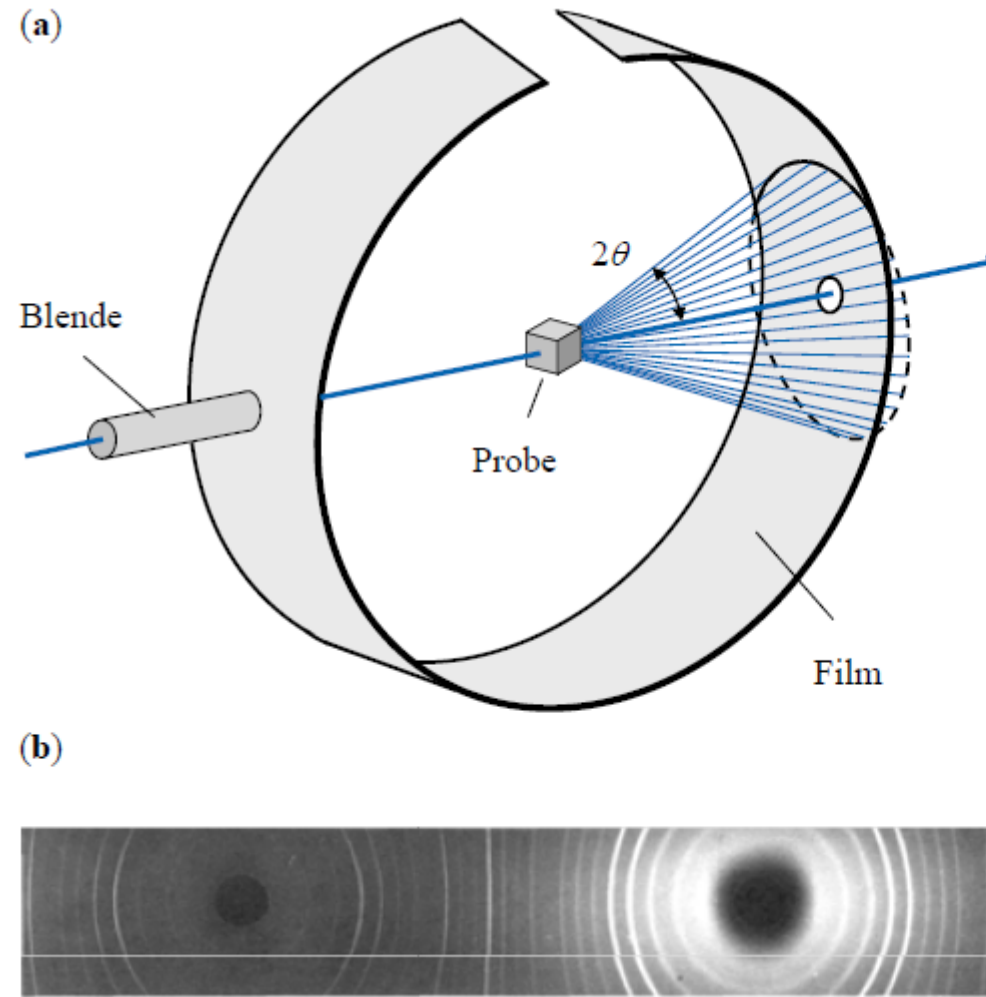


- fcc lattice with 2 atoms in the basis
- $hkl$  all even or all odd
- $hkl$  even  $\rightarrow$  strong
- $hkl$  odd  $\rightarrow$  weak



Hunklinger, Festkörperphysik

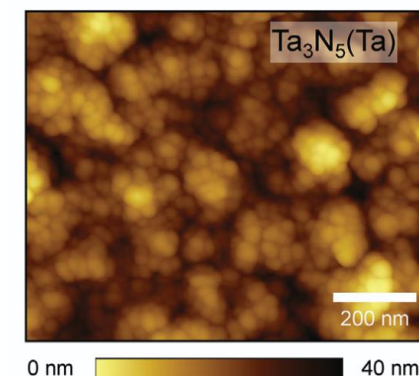
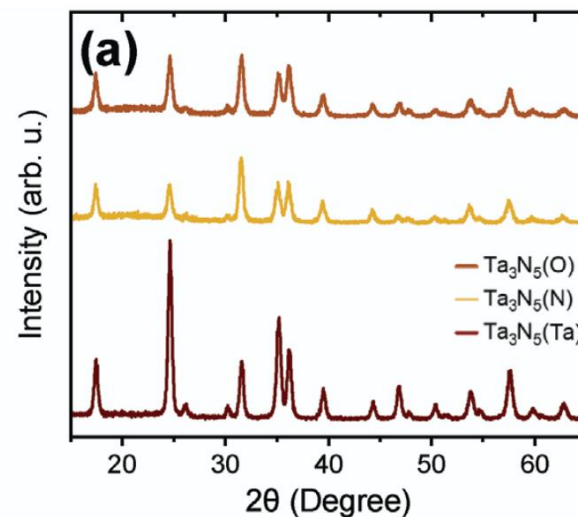
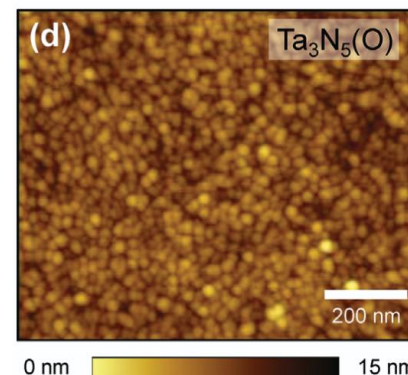
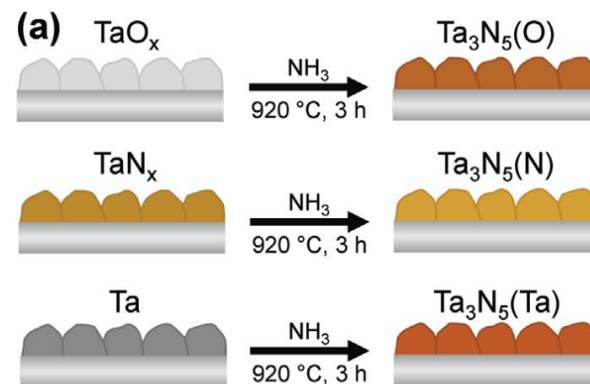
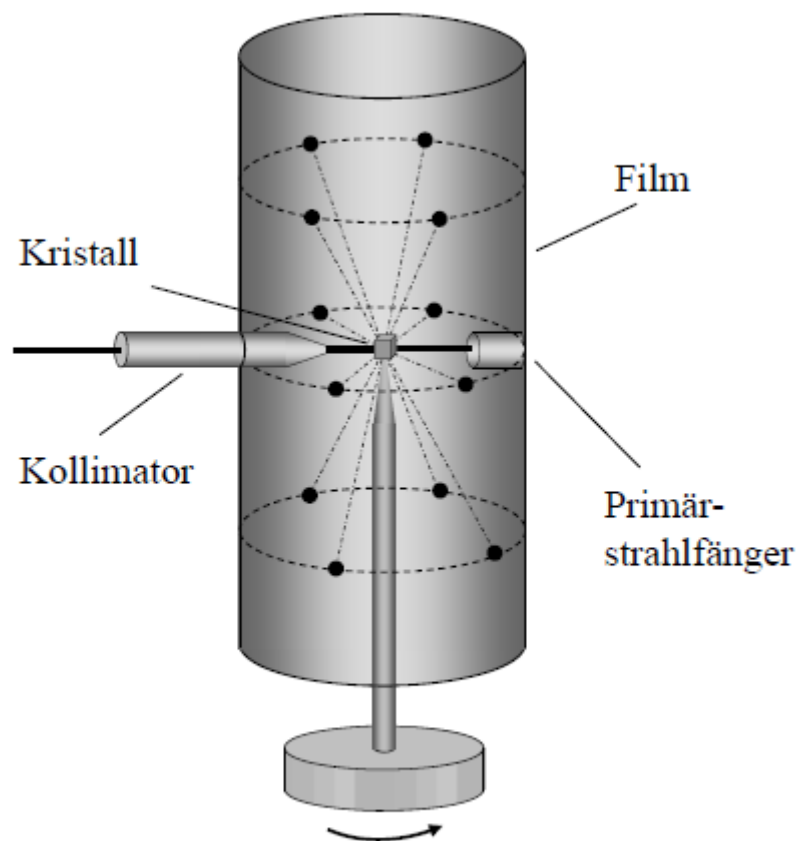
**Bild 4.26:** Laue-Aufnahme eines Lektin-Kristalls. Deutlich zu erkennen ist die zwei-zählige Symmetrie des Kristalls. (Das Bild wurde einem Bericht der CCLRS Synchrotron Radiation Source, Daresbury Laboratories, Großbritannien, entnommen.)



**Bild 4.24:** a) Schematische Darstellung des Debye-Scherrer-Verfahrens. Monochromatische Röntgenstrahlung trifft auf die Probe in der Mitte der Anordnung. Die kegelförmig gestreute Röntgenstrahlung wird durch einen Filmstreifen aufgefangen. b) Pulveraufnahme einer geordneten  $\text{Cu}_3\text{Au}$ -Legierung. Die Beugungsringe wurden wie in der Prinzipskizze mit einem Film registriert. (Nach Ch. Kittel, *Einführung in die Festkörperphysik*, Oldenbourg, 2013.)



Example: semiconductor thin films for energy conversion



Monochromatic X-ray Source



X-ray Detector  
+ precision mechanics

<https://rigaku.com/products/x-ray-diffraction-and-scattering/xrd>